

Microbial Catabolism

Metabolism : total activity of cell

1. Anabolism- used to new molecules
assimilative,
biosynthetic,
energy consuming

2. catabolism - used to obtain energy
degradative,
dissimilative,
energy producing

Overview of Catabolism

Electrons from energy source ☑ redox carriers ☑
terminal electron acceptors

Some free energy released is trapped in
chemically usable forms

Terminal acceptor can be external molecule
(oxygen) in **Respiration** (energy yields high).
***Oxidation of organic compounds uses ETS
(electron transport system) as well as SLP
(substrate level phosphorylation)***

Or terminal acceptor can be organic molecule
derived from energy source during metabolism
(pyruvate) ☑ **Fermentation** (energy yields low)
no oxygen required. Uses substrate level
phosphorylation (SLP)

Carbohydrate Catabolism

Microorganisms oxidize carbohydrates
as their primary source of energy

Glucose - most common energy source

Energy obtained from Glucose by:

Respiration

Fermentation

Aerobic Cellular Respiration

Electrons released by oxidation are passed down an **Electron Transport System** with oxygen being the **Final Electron Acceptor**

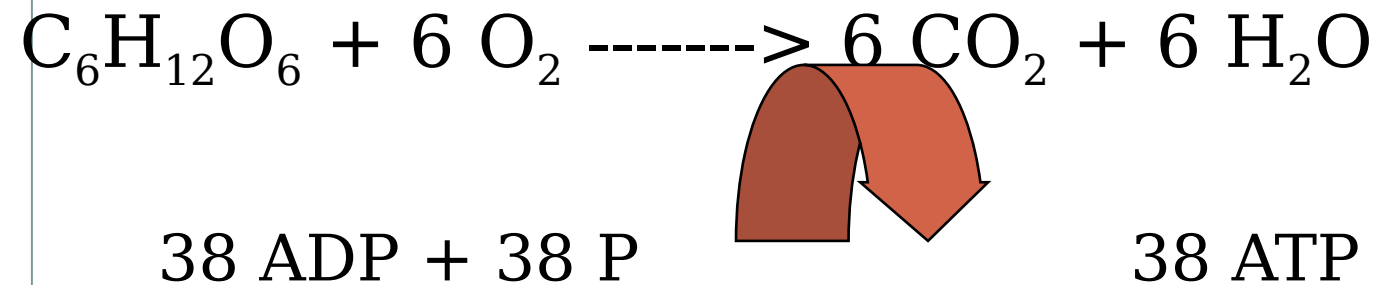
General Equation:

Glucose + oxygen----> Carbon dioxide +
water



ATP

Chemical Equation



Aerobic Cellular Respiration

4 subpathways

1. Glycolysis
2. Transition Reaction
3. Krebs's Cycle
4. Electron Transport System

1. Glycolysis (splitting of sugar)

Oxidation of Glucose into 2 molecules of Pyruvic acid

Embden-Meyerhof Pathway

End Products of Glycolysis:

2 Pyruvic acid

2 NADH_2

2 ATP

2. Transition Reaction

Connects Glycolysis to Krebs Cycle

End Products:

2 Acetyl CoEnzyme A

2 CO₂

2 NADH₂

3. Krebs Cycle (Citric Acid Cycle)

Series of chemical reactions that begin and end with citric acid

Products:

- 2 ATP
- 6 NADH_2
- 2 FADH_2
- 4 CO_2

4. Electron Transport System

Occurs within the cell membrane of
Bacteria

Chemiosmotic Model of Mitchell

34 ATP

How 34 ATP from E.T.S. ?

3 ATP for each NADH_2

2 ATP for each FADH_2

NADH_2

FADH_2

Glycolysis	2
T. R.	2
Krebs Cycle	6
Total	10

Glycolysis	0
T.R.	0
Krebs Cycle	2
Total	2

$10 \times 3 = 30 \text{ ATP}$

$2 \times 2 = 4 \text{ ATP}$

Total ATP production for the complete oxidation of 1 molecule of glucose in Aerobic Respiration

	ATP
Glycolysis	2
Transition Reaction	0
Krebs Cycle	2
E.T.S.	34
Total	38 ATP

- 3. or ***Anaerobic respiration*** (unique to bacteria) oxidation of organic compounds where an external substrate other than oxygen serves as final electron acceptor.
- Eg. Nitrate, sulfate, carbon dioxide

Anaerobic Respiration

- Electrons released by oxidation are passed down an E.T.S., but oxygen **is not** the final electron acceptor
- Nitrate (NO_3^-) ----> Nitrite (NO_2^-)
- Sulfate (SO_4^{2-}) ----> Hydrogen Sulfide (H_2S)
- Carbonate (CO_3^{2-}) -----> Methane (CH_4)

Fermentation

Anaerobic process that does not use the E.T.S. Usually involves the incomplete oxidation of a carbohydrate which then becomes the final electron acceptor.

Glycolysis - plus an additional step

2. Entner-Doudoroff pathway

Another pathway for glucose breakdown discovered by Entner & Doudoroff.

Has evolutionary and taxonomic significance

Found in *Pseudomonas*, *Rhizobium*, *Azotobacter*, almost all gram negatives

Fermentation

Use organic molecules derived from glycolysis being metabolized as electron acceptor.

Pyruvate (or molecules derived by pyruvate) is available from glucose breakdown. Many cells use this as terminal electron acceptor.

Create waste products to be dumped out of cell.

Not as efficient as aerobic respiration; but does allow catabolism to continue in absence of oxygen, better than nothing at all.

Glucose ➡ 2 lactic acid

Some bacteria (lactic acid bacteria, including *Streptococci* and *Lactobacilli*) gain all their energy from fermentation no electron transport chain after.

Lactic acid fermentation

Pyruvate + NADH ➡ Lactic acid + NAD⁺

Enzyme; Lactic acid dehydrogenase

Found in many bacteria, *Bacillus*, also in some protozoa, water molds, even human skeletal muscle

1. Lactic Acid Fermentation

Only 2 ATP

End Product - Lactic Acid

Food Spoilage

Food Production

Yogurt - Milk

Pickles - Cucumbers

Sauerkraut - Cabbage

Genera:

Streptococcus

Lactobacillus

Homolactic fermenters

Some bacteria produce only lactic acid.
Pyruvate is reduced to lactic acid with
the enzyme *lactic acid dehydrogenase*

*Eg. Streptococcus, Enterococcus,
Lactococcus, Lactobacillus sp.*

Collectively called as lactic acid bacteria
*Important in Dairy industry. Responsible
of souring of milk : yogurt, cheese,
buttermilk, sour cream etc. Excellent
keeping properties*

Streptococci living on tooth surface
produce lactic acid.

Lactobacilli in human digestive system
and help digestion of milk

Lactobacillus acidophilus is added to milk
diet of those unable to digest milk

Heterolactic fermenters

some microbes *eg. Leuconostoc* using phosphoketolase pathway instead glycolytic pathway and produce

Ethanol, CO₂ , lactate

Glucose + ADP + pi ➡ Lactic acid + ethanol + CO₂ + ATP

Leuconostoc use in wine production.

Heterolactic fermentars can be used in the fermentation of vegetables eg. Cabbage (sauerkraut) and cucumbers (prickles) and in the manufacture of buttermilk, butter and some cheese.

Alcoholic Fermentation

Pyruvate is decarboxylated to acetaldehyde by enzyme *pyruvate decarboxylase*,

Then reduce to ethanol by *alcohol dehydrogenase*.

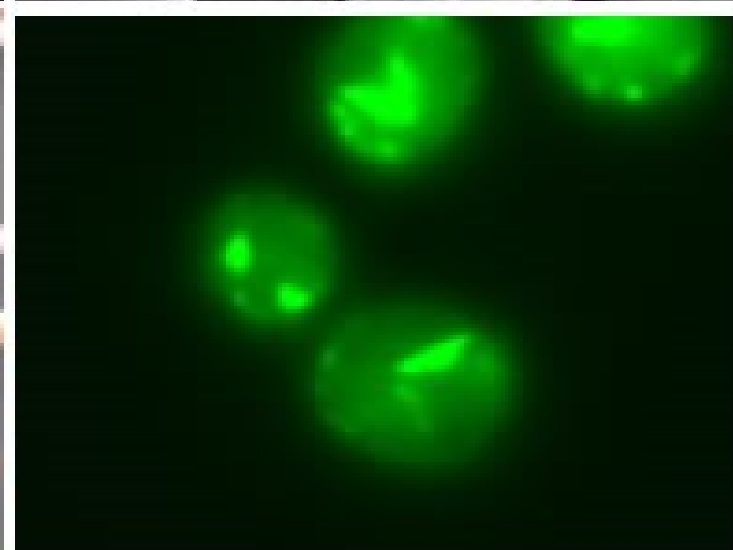
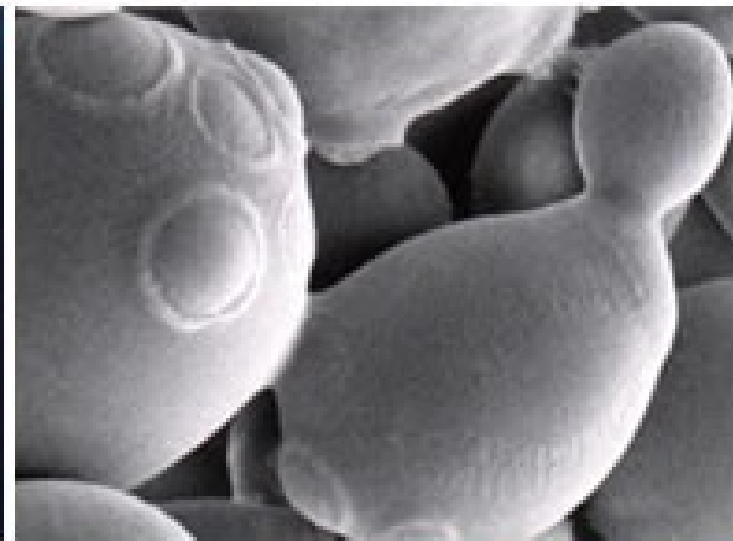
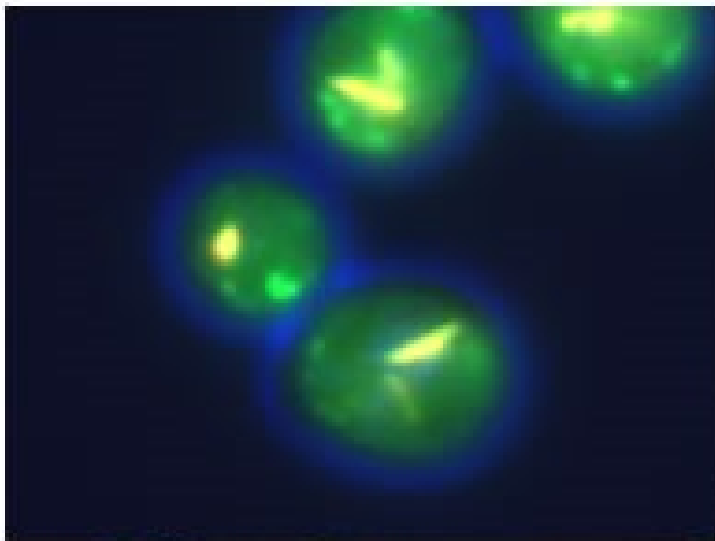
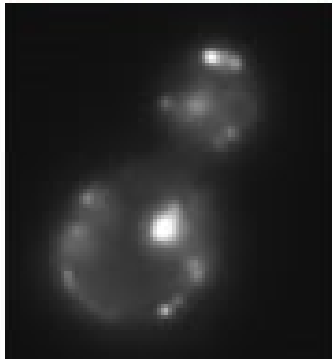
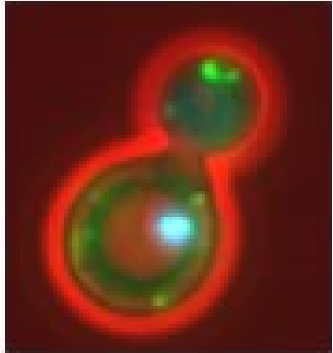
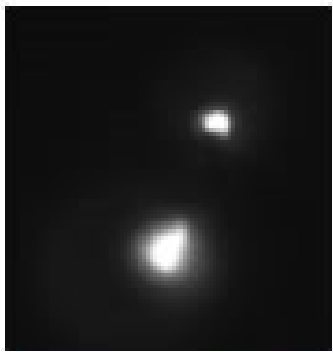
Pyruvate $\rightarrow$ acetaldehyde + CO₂

Acetaldehyde + NADH $\rightarrow$ ethanol + NAD⁺

**Found in few bacteria, in many fungi
yeast**

Yeast *Saccharomyces cerevisiae*

(*Baker's yeast*) use for production of bread. Rising of bread caused by carbon dioxide



***Saccharomyces cerevisiae* is commonly known as baker's, brewer's, or budding yeast. It is a unicellular eukaryotic organism that serves as a model system for studying genomics;**

Fermentation may result in numerous end products



1. Type of organism
2. Original substrate
3. Enzymes that are present and active

Alcohol Fermentation

Only 2 ATP

End products:

alcohol

CO₂

Alcoholic Beverages

Bread dough to rise

Saccharomyces cerevisiae (Yeast)

Mixed acid fermentation

Some bacteria use several pathways produce ethanol, formic acid, acetic acid, lactic acid, succinic acid, CO₂ and H₂

Lots of acid – lower pH than many other fermentations.

Found in the same bacteria of family Enterobacteriaceae

Formic acid fermentation

Pyruvate(3-C) + CoA $\rightarrow$ Acetyl-CoA
(2-C) + formic acid(1-C) - enzyme
formate lyase

$\text{HCOOH} \rightarrow \text{CO}_2 + \text{H}_2$ - enzyme *formate
dehydrogenlyase*

Found in many bacteria, very
common in enterics (gram negative,
facultative anaerobic rods, include
Escherichia coli, *Enterobacter*,
Salmonella, *Proteus* and other.
Useful in identification of enterics.

Propionic Acid Fermentation

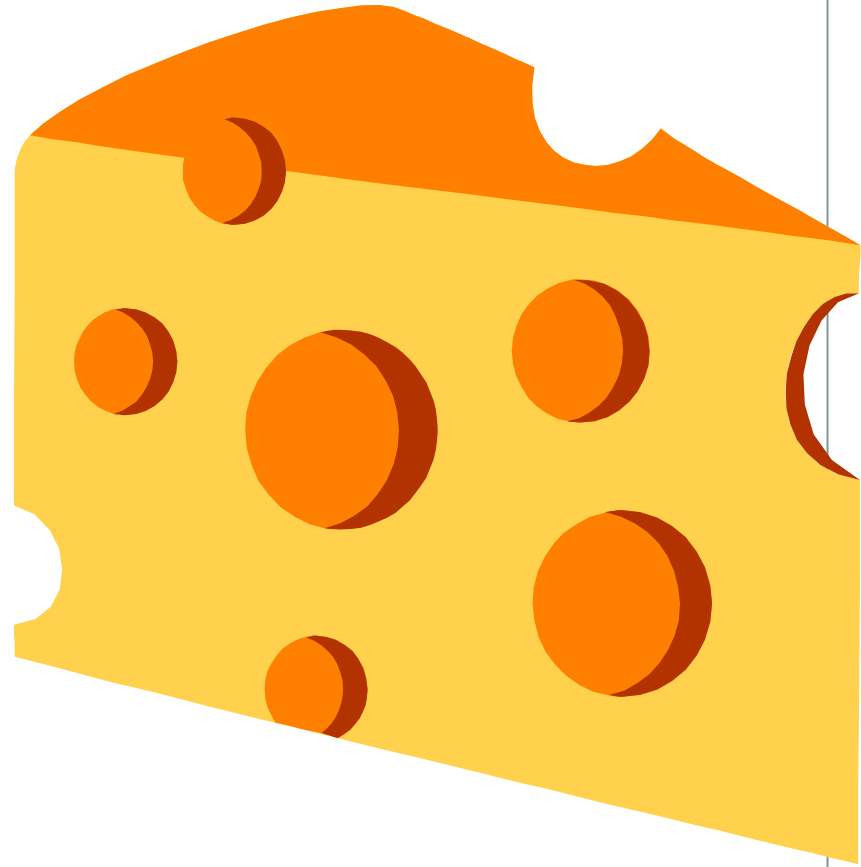
Only 2 ATP

End Products:

Propionic acid

CO₂

Propionibacterium
sp.



Propionic acid fermentation

Genus *Propionibacterium* carry out this fermentation.

Either Glucose or lactic acid converted to propionic acid

Propionic acid fermentation from lactic acid is important in swiss cheese manufacture where fermentative production of CO₂ causes characteristic holes.

Butyric acid fermentation

Pyruvate converted to either
acetone and CO₂

Isopropanol and CO₂

Butanol

Many of these are good organic
solvents.

Species of *Clostridium* carry out
this fermentation.

Differences between respiration in mitochondria (eukaryotes) and bacteria (prokaryotes)

In Eukaryotes:

ETS located in inner mitochondrial membrane, proton gradient develops across inner mitochondrial membrane.

Mitochondria are very efficient at generating proton gradient

With NADH electron donor, P/O ratio can be 3 (means 3 ATP made per NADH)

But with FADH as electron donor, P/O ratio only 2 (fewer protons are transported, less proton gradient)

Overall efficiency of respiration in mitochondria: ~40% (means that about 40% of energy in glucose actually gets converted to ATP)

2. In Prokaryotes

ETS located in cytoplasmic membrane.

Proton gradient develops across this membrane.

Bacteria are not as efficient. ETS chains are shorter. P/O ratios are lower

In prokaryotes P/O ratios for NADH are only ~ 2 . Overall efficiency of glucose oxidation is closer to 28%, not 40%.

TCA cycle (Citric acid cycle/ Krebs cycle)

Formation of acetyl- Co A

Oxidation of pyruvate + NAD^+  Acetyl
CoA + CO_2 + NADH

Carried out by pyruvate dehydrogenase

Net yield: Acetyl CoA + 3NAD^+ + FAD

 2CO_2 + 3NADH + FADH_2 + ATP

TCA cycle completes the oxidation of
carbons in pyruvate to most oxidized
form CO_2

Removes electrons originally in C-H
bonds to electron carriers NADH and
 FADH use in respiration machinery.

Anaerobic respiration:

Use of acceptors other than oxygen.

Most common in bacteria. Most alternative electron acceptors are inorganic molecules, but some organic molecules can serve.

As with aerobic respiration, anaerobic respiration uses ETS, membrane localization, proton gradient, and ATP synthase.

Processes are of great importance both ecologically and industrially.

Examples of anaerobic respiration:

Nitrate (NO_3^-).

Process called *denitrification*- called dissimilative nitrate reduction

Organisms respiring anaerobically gain less energy than with oxygen

Enzyme *nitrate reductase*. Enzyme is repressed by oxygen, synthesis turned on in absence of oxygen.

Process can have several steps, proceed in two different directions

1. nitrate  nitrite    ammonia
2. nitrate  nitrite  nitrous oxide
(N₂O)   dinitrogen gas

Second process is major pathway for loss of nitrogen compounds from soil, return of nitrogen to atmosphere

***Pseudomonas* species are common denitrifiers, widespread in soils. When fertilized soils become flooded, oxygen is rapidly depleted, *Pseudomonas* switch to anaerobic respiration and leaving field in unfertile state**

2. Sulfate (SO_4^{2-}).

Process called **sulfate reduction**.

Sulfate (SO_4^{2-}) $\rightarrow$ $\rightarrow$ $\rightarrow$ $\rightarrow$ Hydrogen Sulfide (H_2S)

Small group of bacteria carry out this reaction; all obligate anaerobes.

Have unique cytochrome c3.

Sulfate is common in seawater. Often, H_2S combines with iron, forms insoluble FeS $\rightarrow$ black sediments. Common in estuaries.

3. Carbon dioxide (CO₂).

One of most common inorganic ions.

Methanogens: most important group of CO₂ reducers. Obligate anaerobes, archaeobacteria. Produce methane as waste product.

Reaction: $\text{CO}_2 + \text{H}_2 + \text{H}^+ \rightarrow \text{CH}_4 + \text{H}_2\text{O}$

Note: reaction also requires Hydrogen gas. Methanogens typically live alongside bacteria that produce hydrogen by fermentation, remove hydrogen as it is made.

Catabolism of substances other than glucose

Many other possible C-sources for catabolism beside glucose. In general, must convert these into molecules that can enter into central metabolism, either in glycolysis or TCA cycle.

1. Carbohydrates

Most abundant C-sources in most environments, most in various polysaccharides (cellulose, starch, lignin, etc.)

To gain access to sugars, must first secrete hydrolytic enzymes that break down glycosidic bonds in polysaccharides, produce mono- and disaccharides that can be transported into cells.

Starch, glycogen -- easily hydrolyzed by amylases

Cellulose -- difficult to digest, very insoluble, tightly folded. Many fungi, some bacteria produce cellulases.

2.lipids

Biological lipids common as triglycerides, diglycerides.

To catabolize, bacteria secrete lipases, hydrolyze glycerides to free fatty acids and glycerol.

Fatty acids attacked by Beta-oxidation pathway.

Using FAD and NAD⁺ to remove electrons, 2-C units are removed as Acetyl-CoA, feed directly into central metabolism at TCA cycle entry. Glycolysis pathway not involved.

Mixed amino acid fermentation

Some microbes can ferment substances other than carbohydrate.

Genus *Clostridium* (eg. *C. sporogenes*, *C. botulinum*) can ferment mixtures of aminoacids

This process is called **Stickland reaction**

Clostridium (obligate anaerobes): can oxidize pyruvate to acetyl Co-A.

Hydrogen gas can be liberated directly

Responsible for gas produced by botulism (*C. botulinum*) that causes swollen cans in improperly sterilized canning process.

3.proteins

Proteins must first be hydrolyzed by protease enzymes, to get individual amino acids which can be transported into cells.

Amino acids all have common structure:
 $\text{NH}_2 - \text{RCH} - \text{COOH}$.

1st step in catabolism is to remove amino group (deamination), often by swapping it with another substrate (transamination).

Typical example: glutamic acid (an AA)
+ pyruvate $\rightarrow$ alpha-ketoglutarate +
alanine (= pyruvate + amino group).
Now alpha-KG can be oxidized in TCA
cycle, since it is a TCA cycle
compound.

As excess amino groups accumulate,
must be secreted as waste products,
possibly as ammonium ion (leads to
alkaline pH).